

LEC10 - Kepler's & Newton's Laws

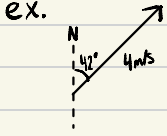
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Learning Goals:

- Explain Kepler's laws and how they describe planetary orbits in the Solar System
- Apply Newton's laws and the law of gravitation to explain why and how objects orbit due to gravity

Speed: The magnitude of a velocity vector

Velocity: A vector value involving a speed
ex. 3m/s[North]
ex. -4.3km/h[SE]



Review of Kepler's Laws:

Kepler's First Law → Planetary orbits are ellipses

Kepler's Second Law → Planets sweep out equal areas in equal times in their orbits

Kepler's Third Law → Planetary orbits obey the relationship $p^2 = a^3$ (In our Solar System only)

UNITS MATTER (YEARS & AU)

Newton's Laws of Motion and Gravity:

1. An object will remain at a constant velocity unless acted on by a net force

- **Speed** → The rate at which an object moves (distance per time)

- **Velocity** → Speed and direction (Vector value)

- **Force** → A push or pull that causes acceleration

2. Net force = mass × acceleration ($F=ma$)

- **Net Force** → Sum of all forces acting on a body (1 Newton of force = $\text{kg} \cdot \text{m/s}^2$)

- **Mass** → Amount of atoms in an object

- **Weight** is the force that an object experiences (Expressed in pounds, Newtons, etc.)

- **Acceleration** → Change in magnitude or direction of velocity (m/s^2)

- If the only force acting on an object is gravity, it is in free fall

- The acceleration due to gravity is called g . On Earth, its value is $\approx 10\text{m/s}^2$

- For free-falling objects, Newton's 2nd law becomes: $F_g = mg$

3. Every force causes an "equal and opposite" reaction force

- The Earth pulls on a falling ball, and the ball pulls on the Earth equally hard

- A jet engine is pushed forward, while the exhaust flow is pushed backwards

Gravity:

- The Universal Law of Gravitation tells us the strength of the gravitational attraction between two objects

$$F_g = G \frac{m_1 m_2}{d^2}$$

Force of Gravity (N) Gravitational Constant Masses of the 2 objects (kg) Distance between the CENTERS of the 2 objects

- The distance (d) between the Earth vs. an astronaut on the ISS

- From surface to ISS → 6000km → 664N

- From center of Earth to ISS → 6000km + 400km (Radius of Earth) → 584N

- Astronauts feel weightless in space because they are in a constant free fall

- Centripetal motion keeps the Astronaut from hitting Earth

* The reason Kepler's third law works well without $(m_1 + m_2)$ is that the sum is very close to 1 (ex. Earth is $\sim 0.000035M$)

Newton's Version of Kepler's Third Law (In a general orbiting system including but not limited to the Solar System):

$$p^2 = \frac{a^3}{(M_1 + M_2)}$$

Orbital Period (years) Average orbital distance between centers (AU) Sum of object masses in Solar Masses

• Review of Kepler's 3 laws from LEC9

• Newton's law of universal gravitation allows us to calculate the force of gravitational attraction between any 2 objects

• Newton derived the exact formula for Kepler's Third Law, which lets us include any orbiting system because it accounts for the masses of the two planetary bodies